# 27 V ESD Protection Diode
## Dual Line CAN Bus Protector

# NUP2105L, SZNUP2105L

The SZ/NUP2105L has been designed to protect the CAN transceiver in high−speed and fault tolerant networks from ESD and other harmful transient voltage events. This device provides bidirectional protection for each data line with a single compact SOT−23 package, giving the system designer a low cost option for improving system reliability and meeting stringent EMI requirements.

### Features

- 350 W Peak Power Dissipation per Line (8/20 μsec Waveform)
- Low Reverse Leakage Current (< 100 nA)
- Low Capacitance High−Speed CAN Data Rates
- IEC Compatibility: – IEC 61000−4−2 (ESD): Level 4, 30 kV
  – IEC 61000−4−4 (EFT): 40 A – 5/50 ns
  – IEC 61000−4−5 (Lighting) 8.0 A (8/20 μs)
- ISO 7637−2 Pulse 2a: Repetitive Load Switch Disconnect, 9.5 A
- ISO 7637−3 Pulse 3a,b: Repetitive Load Switching Fast Transients, 50 A
- Flammability Rating UL 94 V−0
- SZ Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

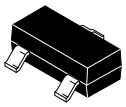
### Applications

- Industrial Control Networks
  - Smart Distribution Systems (SDS®)
  - DeviceNet™
- Automotive Networks
  - Low and High−Speed CAN
  - Fault Tolerant CAN

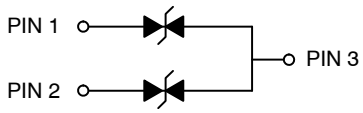
**ON Semiconductor®**

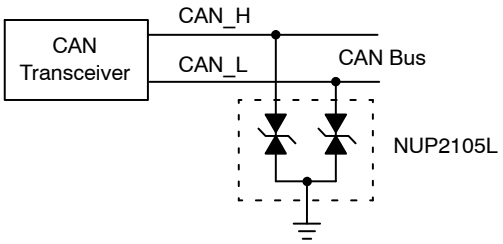
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**SOT−23**
**DUAL BIDIRECTIONAL**
**VOLTAGE SUPPRESSOR**
**350 W PEAK POWER**

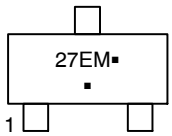


**SOT−23**
**CASE 318**
**STYLE 28**




**MARKING DIAGRAM**




27E = Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

**ORDERING INFORMATION**

See detailed ordering and shipping information in the package dimensions section on page 2 of this data sheet.

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**MAXIMUM RATINGS** ($T_J$ = 25°C, unless otherwise specified)

| Symbol | Rating | Value | Unit |
|---|---|---|---|
| PPK | Peak Power Dissipation<br>8/20 μs Double Exponential Waveform (Note 1) | <br>350 | W |
| $T_J$ | Operating Junction Temperature Range | −55 to 150 | °C |
| $T_J$ | Storage Temperature Range | −55 to 150 | °C |
| $T_L$ | Lead Solder Temperature (10 s) | 260 | °C |
| ESD | Human Body model (HBM)<br>Machine Model (MM)<br>IEC 61000−4−2 Specification (Contact) | 16<br>400<br>30 | kV<br>V<br>kV |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Non−repetitive current pulse per Figure 1.

**ELECTRICAL CHARACTERISTICS** ($T_J$ = 25°C, unless otherwise specified)

| Symbol | Parameter | Test Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $V_{RWM}$ | Reverse Working Voltage | (Note 2) | 24 | – | – | V |
| $V_{BR}$ | Breakdown Voltage | $I_T$ = 1 mA (Note 3) | 26.2 | – | 32 | V |
| $I_R$ | Reverse Leakage Current | $V_{RWM}$ = 24 V | – | 1.5 | 100 | nA |
| $V_C$ | Clamping Voltage | $I_{PP}$ = 5 A (8/20 μs Waveform)<br>(Note 4) | – | – | 40 | V |
| $V_C$ | Clamping Voltage | $I_{PP}$ = 8 A (8/20 μs Waveform)<br>(Note 4) | – | – | 44 | V |
| $I_{PP}$ | Maximum Peak Pulse Current | 8/20 μs Waveform (Note 4) | – | – | 8.0 | A |
| CJ | Capacitance | $V_R$ = 0 V, f = 1 MHz (Line to GND) | – | – | 30 | pF |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Surge protection devices are normally selected according to the working peak reverse voltage ($V_{RWM}$), which should be equal or greater than the DC or continuous peak operating voltage level.
3. $V_{BR}$ is measured at pulse test current $I_T$.
4. Pulse waveform per Figure 1.

**ORDERING INFORMATION**

| Device | Package | Shipping† |
|---|---|---|
| NUP2105LT1G | SOT−23<br>(Pb−Free) | 3,000 / Tape & Reel |
| SZNUP2105LT1G* | SOT−23<br>(Pb−Free) | 3,000 / Tape & Reel |
| NUP2105LT3G | SOT−23<br>(Pb−Free) | 10,000 / Tape & Reel |
| SZNUP2105LT3G* | SOT−23<br>(Pb−Free) | 10,000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

*SZ Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable

## TYPICAL PERFORMANCE CURVES

($T_J$ = 25°C unless otherwise noted)

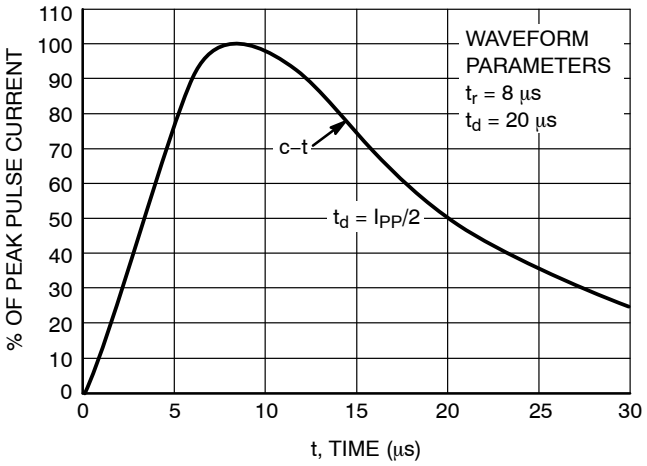



**Figure 1. Pulse Waveform, IEC 61000−4−5 8/20 μs**

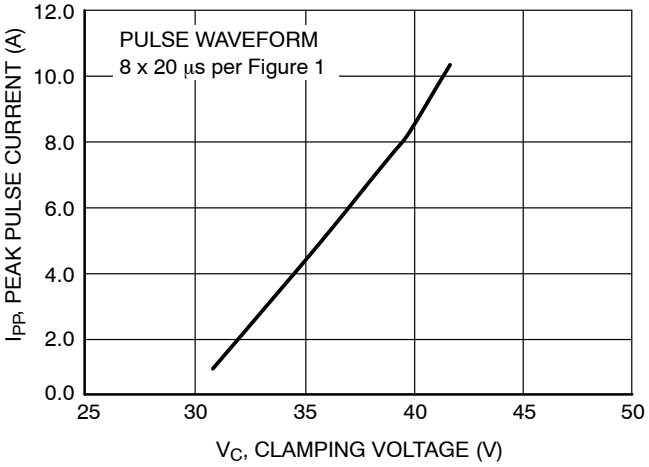



**Figure 2. Clamping Voltage vs Peak Pulse Current**

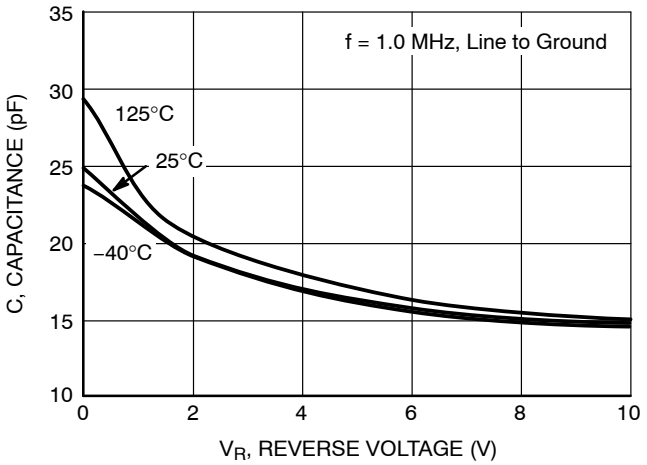



**Figure 3. Typical Junction Capacitance vs Reverse Voltage**

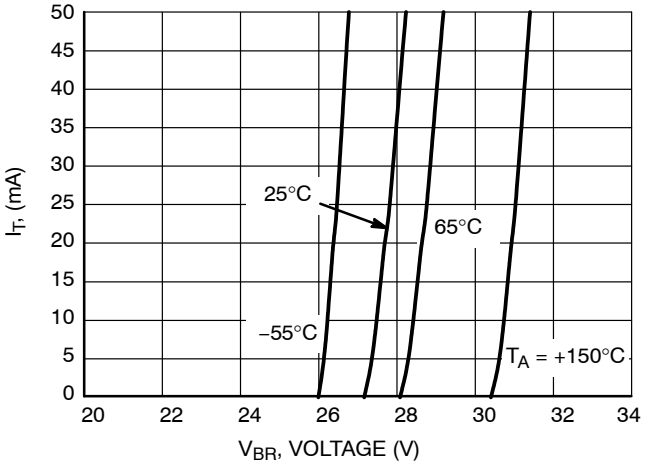



**Figure 4. $V_{BR}$ versus $I_T$ Characteristics**

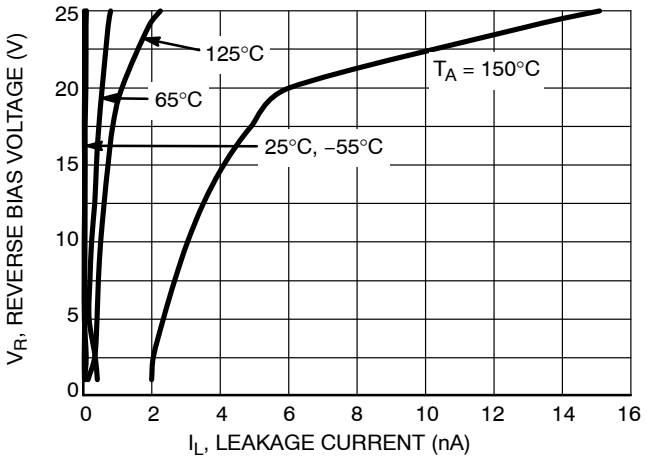



**Figure 5. $I_R$ versus Temperature Characteristics**

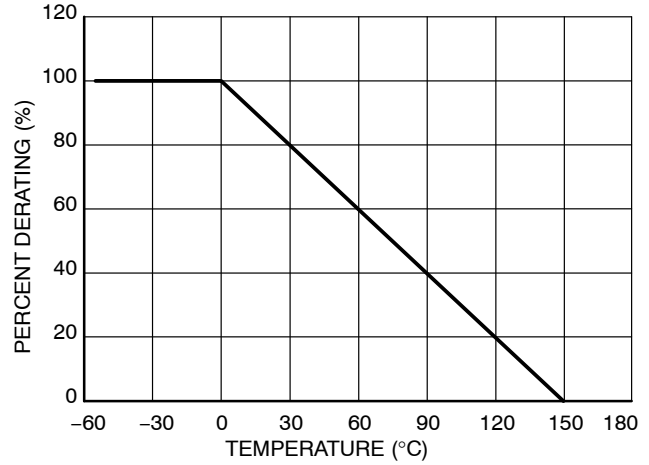



**Figure 6. Temperature Power Dissipation Derating**

## APPLICATIONS

### Background

The Controller Area Network (CAN) is a serial communication protocol designed for providing reliable high speed data transmission in harsh environments. surge protection diodes provide a low cost solution to conducted and radiated Electromagnetic Interference (EMI) and Electrostatic Discharge (ESD) noise problems. The noise immunity level and reliability of CAN transceivers can be easily increased by adding external surge protection diodes to prevent transient voltage failures.

The NUP2105L provides a surge protection solution for CAN data communication lines. The NUP2105L is a dual bidirectional surge protection device in a compact SOT−23 package. This device is based on Zener technology that optimizes the active area of a PN junction to provide robust protection against transient EMI surge voltage and ESD. The NUP2105L has been tested to EMI and ESD levels that exceed the specifications of popular high speed CAN networks.

### CAN Physical Layer Requirements

Table 1 provides a summary of the system requirements for a CAN transceiver. The ISO 11898−2 physical layer specification forms the baseline for most CAN systems. The transceiver requirements for the Honeywell® Smart Distribution Systems (SDS®) and Rockwell (Allen−Bradley) DeviceNet™ high speed CAN networks are similar to ISO 11898−2. The SDS and DeviceNet transceiver requirements are similar to ISO 11898−2; however, they include minor modifications required in an industrial environment.

**Table 1. Transceiver Requirements for High−Speed CAN Networks**

| **Parameter** | ISO 11898−2 | SDS Physical Layer Specification 2.0 | DeviceNet |
|---|---|---|---|
| **Min / Max Bus Voltage (12 V System)** | −3.0 V / 16 V | 11 V / 25 V | Same as ISO 11898−2 |
| **Common Mode Bus Voltage** | CAN_L:<br>−2.0 V (min)<br>2.5 V (nom)<br>CAN_H:<br>2.5 V (nom)<br>7.0 V (max) | Same as ISO 11898−2 | Same as ISO 11898−2 |
| **Transmission Speed** | 1.0 Mb/s @ 40 m<br>125 kb/s @ 500 m | Same as ISO 11898−2 | 500 kb/s @ 100 m<br>125 kb/s @ 500 m |
| **ESD** | Not specified, recommended ≥ ±8.0 kV (contact) | Not specified, recommended ≥ ±8.0 kV (contact) | Not specified, recommended ≥ ±8.0 kV (contact) |
| **EMI Immunity** | ISO 7637−3, pulses ‘a’ and ‘b’ | IEC 61000−4−4 EFT | Same as ISO 11898−2 |
| **Popular Applications** | Automotive, Truck, Medical and Marine Systems | Industrial Control Systems | Industrial Control Systems |

### EMI Specifications

The EMI protection level provided by the surge protection device can be measured using the International Organization for Standardization (ISO) 7637−2 and −3 specifications that are representative of various noise sources. The ISO 7637−2 specification is used to define the susceptibility to coupled transient noise on a 12 V power supply, while ISO 7637−3 defines the noise immunity tests for data lines. The ISO 7637 tests also verify the robustness and reliability of a design by applying the surge voltage for extended durations.

The IEC 61000−4−X specifications can also be used to quantify the EMI immunity level of a CAN system. The IEC 61000−4 and ISO 7637 tests are similar; however, the IEC standard was created as a generic test for any electronic system, while the ISO 7637 standard was designed for vehicular applications. The IEC61000−4−4 Electrical Fast Transient (EFT) specification is similar to the ISO 7637−3 pulse 3a and b tests and is a requirement of SDS CAN systems. The IEC 61000−4−5 test is used to define the power absorption capacity of a surge protection device and long duration voltage transients such as lightning. Table 2 provides a summary of the ISO 7637 and IEC 61000−4−X test specifications. Table 3 provides the NUP2105L's ESD test results.

**Table 2. ISO 7637 and IEC 61000−4−X Test Specifications**

| Test | Waveform | Test Specifications | NUP2105L Results | Simulated Noise Source |
|---|---|---|---|---|
| ISO 7637−2<br>12 V Power Supply Lines (Note 2) | Pulse 1 | $V_s$ = 0 to −100 V<br>$I_{max}$ = 10 A<br>$t_{duration}$ = 5000 pulses | $I_{max}$ = 1.75 A<br>$V_{clamp_max}$ = 31 V<br>$t_{duration}$ = 5000 pulses<br>$R_i$ = 10 Ω, $t_r$ = 1.0 μs,<br>$t_{d_10\%}$ = 2000 μs, $t_1$ = 2.5 s,<br>$t_2$ = 200 ms, $t_3$ = 100 μs | DUT (Note 1) in parallel with inductive load that is disconnected from power supply. |
| | Pulse 2a | $V_s$ = 0 to +50 V<br>coupled onto 14 V battery<br>$I_{max}$ = 10 A<br>$t_{duration}$ = 5000 pulses | $I_{max}$ = 9.5 A<br>$V_{clamp_max}$ = 42 V<br>$t_{duration}$ = 5000 pulses<br>Ri = 2 Ω, $t_r$ = 1.0 μs,<br>$t_{d_10\%}$ = 50 μs, $t_1$ = 2.5 s,<br>$t_2$ = 200 ms | DUT in series with inductor (wire harness) that is disconnected from load. |
| ISO 7637−3<br>Repetitive data line fast transients (Note 3) | Pulse 'a' | $V_s$ = −60 V<br>$I_{max}$ = 1.2 A<br>$t_{duration}$ = 10 minutes | $I_{max}$ = 50 A (Note 4)<br>$V_{clamp_max}$ = 40 V<br>$t_{duration}$ = 60 minutes<br>$R_i$ = 50 Ω, $t_r$ = 5.0 ns,<br>$t_{d_10\%}$ = 100 ns, $t_1$ = 100 μs,<br>$t_2$ = 10 ms, $t_3$ = 90 ms | Switching noise of inductive loads. |
| | Pulse 'b' | $V_s$ = +40 V<br>$I_{max}$ = 0.8 A<br>$t_{duration}$ = 10 minutes | | |
| IEC 61000−4−4<br>Data Line EFT | | $V_{open\ circuit}$ = 2.0 kV<br>$I_{short\ circuit}$ = 40 A<br>(Level 4 = Severe Industrial Environment)<br>$R_i$ = 50 Ω, $t_r$ < 5.0 ns,<br>$t_{d_50\%}$ = 50 ns, $t_{burst}$ = 15 ms,<br>$f_{burst}$ = 2.0 to 5.0 kHz,<br>$t_{repeat}$ = 300 ms<br>$t_{duration}$ = 1 minute | (Note 5) | Switching noise of inductive loads. |
| IEC 61000−4−5 | | $V_{open\ circuit}$ = 1.2/50 μs,<br>$I_{short\ circuit}$ = 8/20 μs<br>$R_i$ = 50 Ω | $I_{max}$ = 8.0 A | Lightning, nonrepetitive power line and load switching |

1. DUT = device under test.
2. Test specifications were taken from ISO7637−2: 2004 version.
3. Test specifications were taken from ISO7637−3: 1995 version.
4. DUT was tested to ISO7637−2: 2004 pulse 3a,b specification for more rigorous test.
5. The EFT immunity level was measured with test limits beyond the IEC 61000−4−4 test, but with the more severe test conditions of ISO 7637−3.

**Table 3. NUP2105L ESD Test Results**

| ESD Specification | Test | Test Level | Pass / Fail |
|---|---|---|---|
| Human Body Model | Contact | 16 kV | Pass |
| IEC 61000−4−2 | Contact | 30 kV (Note 6) | Pass |
| | Non−contact (Air Discharge) | 30 kV (Note 6) | Pass |

6. Test equipment maximum test voltage is 30 kV.

### Surge protection Diode Protection Circuit

surge protection diodes provide protection to a transceiver by clamping a surge voltage to a safe level. surge protection diodes have high impedance below and low impedance above their breakdown voltage. A surge protection Zener diode has its junction optimized to absorb the high peak energy of a transient event, while a standard Zener diode is designed and specified to clamp a steady state voltage.

Figure 7 provides an example of a dual bidirectional surge protection diode array that can be used for protection with the high−speed CAN network. The bidirectional array is created from four identical Zener surge protection diodes. The clamping voltage of the composite device is equal to the breakdown voltage of the diode that is reversed biased, plus the diode drop of the second diode that is forwarded biased.

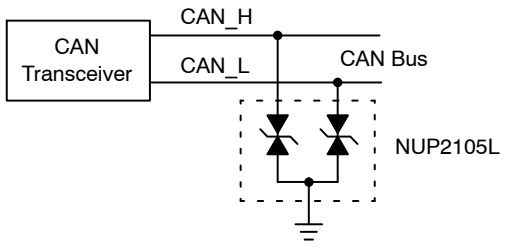
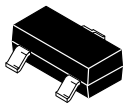



**Figure 7. High−Speed and Fault Tolerant CAN Surge Protection Circuit**

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DeviceNet is a trademark of Rockwell Automation.

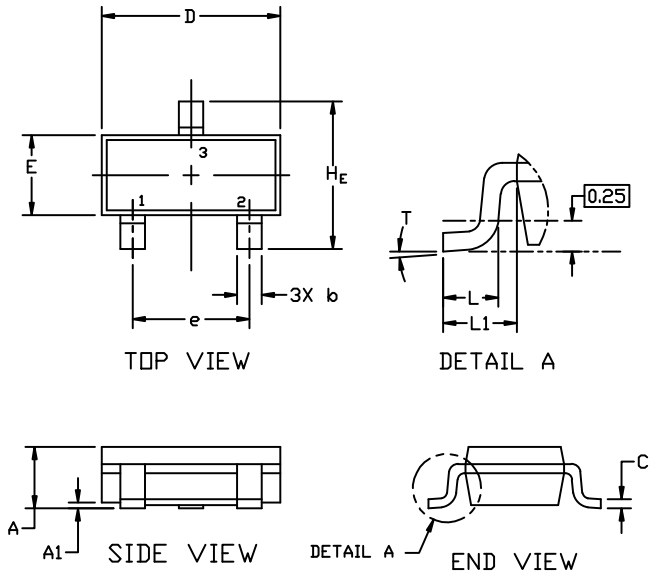
# MECHANICAL CASE OUTLINE
## PACKAGE DIMENSIONS



SCALE 4:1

**SOT−23 (TO−236)**
CASE 318
ISSUE AT

DATE 01 MAR 2023

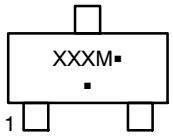



NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M,1994.
2. CONTROLLING DIMENSION: MILLIMETERS
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

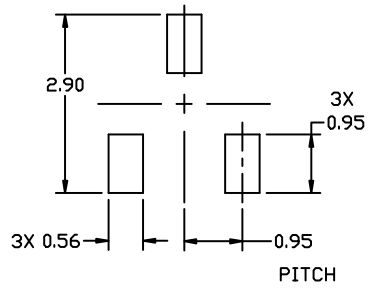
| | MILLIMETERS | | | INCHES | | |
|---|---|---|---|---|---|---|
| DIM | MIN. | NOM. | MAX. | MIN. | NOM. | MAX. |
| A | 0.89 | 1.00 | 1.11 | 0.035 | 0.039 | 0.044 |
| A1 | 0.01 | 0.06 | 0.10 | 0.000 | 0.002 | 0.004 |
| b | 0.37 | 0.44 | 0.50 | 0.015 | 0.017 | 0.020 |
| c | 0.08 | 0.14 | 0.20 | 0.003 | 0.006 | 0.008 |
| D | 2.80 | 2.90 | 3.04 | 0.110 | 0.114 | 0.120 |
| E | 1.20 | 1.30 | 1.40 | 0.047 | 0.051 | 0.055 |
| e | 1.78 | 1.90 | 2.04 | 0.070 | 0.075 | 0.080 |
| L | 0.30 | 0.43 | 0.55 | 0.012 | 0.017 | 0.022 |
| L1 | 0.35 | 0.54 | 0.69 | 0.014 | 0.021 | 0.027 |
| $H_E$ | 2.10 | 2.40 | 2.64 | 0.083 | 0.094 | 0.104 |
| T | 0° | --- | 10° | 0° | --- | 10° |

### GENERIC MARKING DIAGRAM*




XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.




RECOMMENDED MOUNTING FOOTPRINT

* For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**STYLES ON PAGE 2**

| | | | |
|---|---|---|---|
| **DOCUMENT NUMBER:** | **98ASB42226B** | Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped “CONTROLLED COPY” in red. | |
| **DESCRIPTION:** | **SOT−23 (TO−236)** | | **PAGE 1 OF 2** |

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**SOT−23 (TO−236)**
CASE 318
ISSUE AT

DATE 01 MAR 2023

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE



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